

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\FE\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'NG_FFCutsFits.mat');

if ~exist('NG_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    NG_FFCutsFits = ld.fit_src_struct;
end

```

```

fn = fieldnames(NG_FFCutsFits);
nf = 11

```

```

nf =
11

```

```

%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei400_off110NGminFF'

```

```

selectCuts =
'Ei400_off110NGminFF'

```

```

if contains(selectCuts, 'msk')
    error('Can not fit field: %s', selectCuts);
end
if contains(selectCuts, 'all_fit_par')
    error('Can not fit field: %s', selectCuts);
end

fprintf('Processing cuts: %s \n', selectCuts);

```

```

Processing cuts: Ei400_off110NGminFF

```

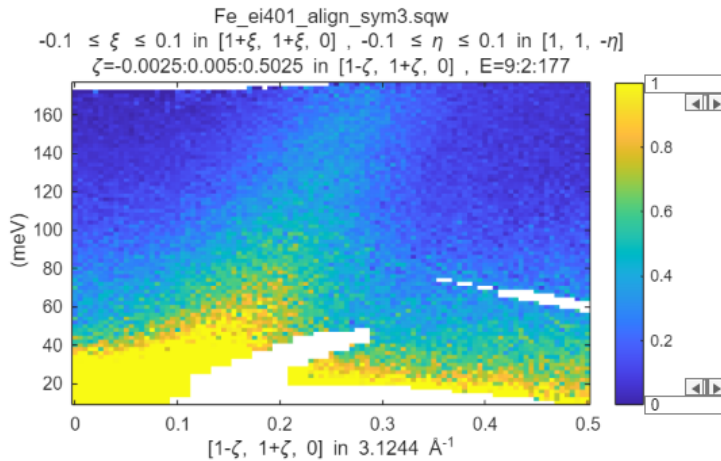
```

%selectCuts= 'Ei400_off200GPmaxFF';
cuts_list = NG_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts, 'Ei200')
    mi = maps_instrument(200, 600, 'S');
end
if contains(selectCuts, 'Ei400')
    mi = maps_instrument(401, 600, 'S');
end
if contains(selectCuts, 'Ei800')
    mi = maps_instrument(786, 600, 'S');
end
sample=IX_sample(true, [1,0,0], [0,1,0], 'cuboid', [0.04,0.03,0.02]);
for j=1:numel(cuts_list)

```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "NG";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

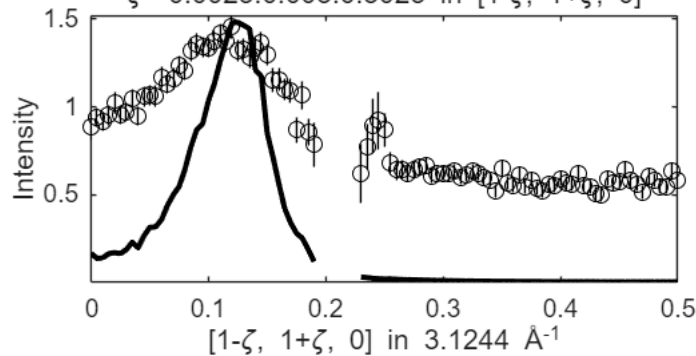
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
NG_FFCutsFits.(selectCuts) = cuts_list;

NG_FFCutsFits = fit_multicutsFF_along_direction(...
    NG_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

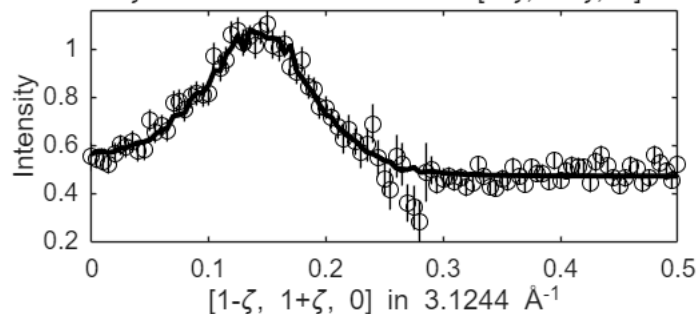
```
*****
**** En = 35±5 going to 165
*****
```

Fe_ei401_align_sym3.sqw
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, $3(\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

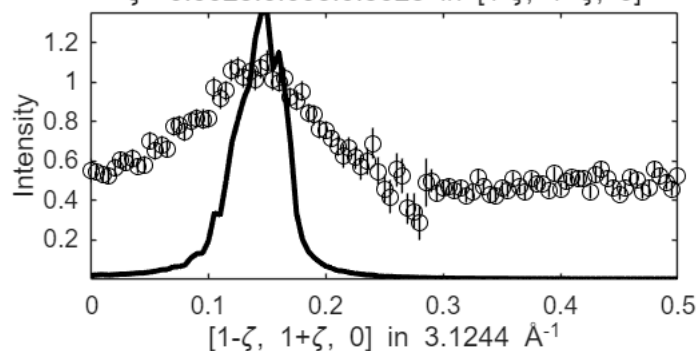


 **** En = 45±5 going to 165

Fe_ei401_align_sym3.sqw
 1 ± 0.014 ; $J_0 = 23 \pm 0.3$; $J_1 = 0 \pm 0$; $\gamma = 4.0407 \pm 0.1$
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, $4(\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

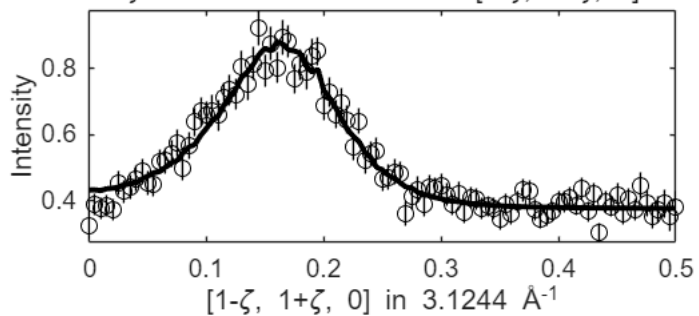


Fe_ei401_align_sym3.sqw
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, $4(\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

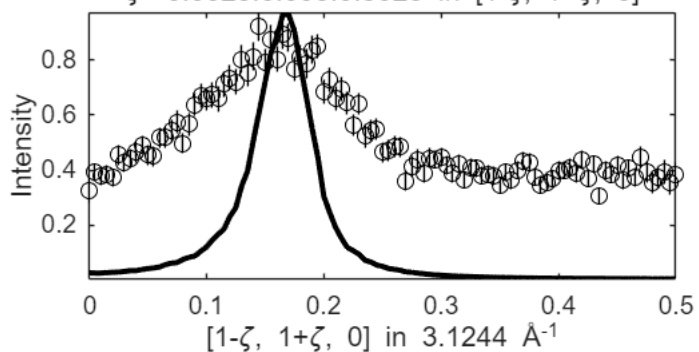


 **** En = 55±5 going to 165

Fe_ei401_align_sym3.sqw
 7 ± 0.015 ; $J_0 = 24 \pm 0.32$; $J_1 = 0 \pm 0$; $\gamma = 9.81346 \pm 0.0001$
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, $5(1-\zeta) \leq \zeta \leq 5(1+\zeta)$ in $[1-\zeta, 1+\zeta, 0]$

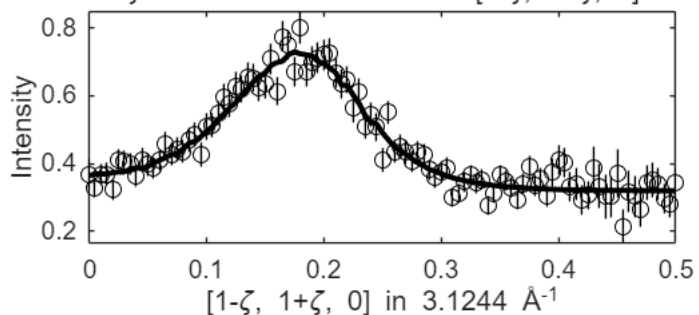


Fe_ei401_align_sym3.sqw
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, $5(1-\zeta) \leq \zeta \leq 5(1+\zeta)$ in $[1-\zeta, 1+\zeta, 0]$

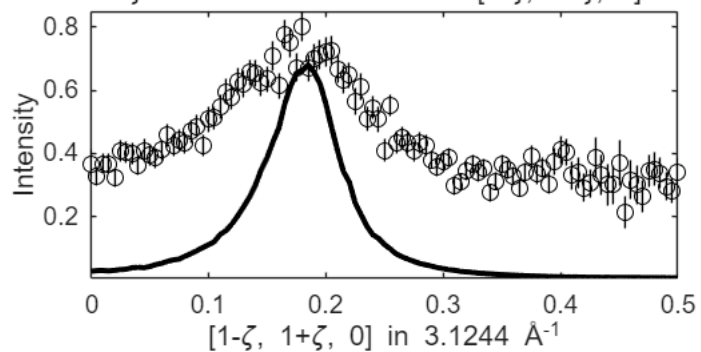


 **** En = 65 ± 5 going to 165

Fe_ei401_align_sym3.sqw
 6 ± 0.019 ; $J_0 = 25 \pm 0.32$; $J_1 = 0 \pm 0$; $\gamma = 18.186 \pm 0.0001$
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, $6(1-\zeta) \leq \zeta \leq 6(1+\zeta)$ in $[1-\zeta, 1+\zeta, 0]$

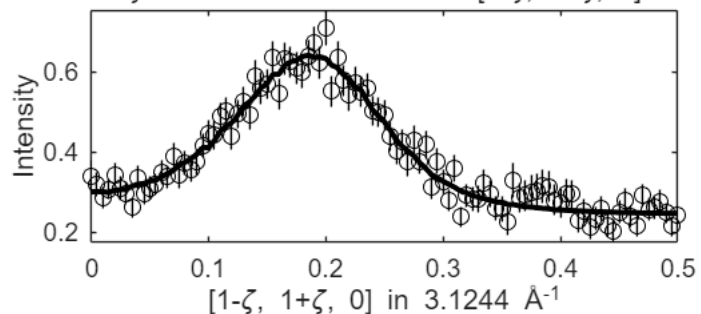


Fe_ei401_align_sym3.sqw
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 6σ
 $\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

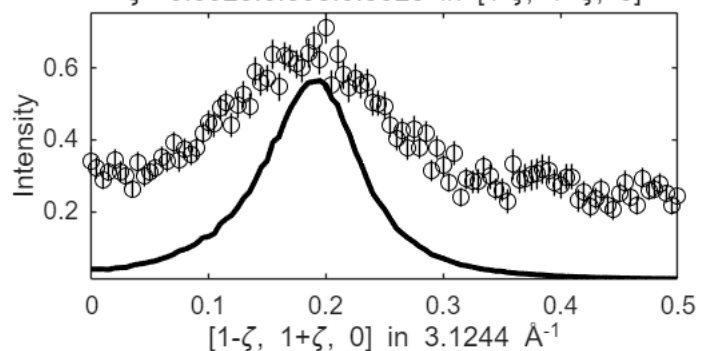


 **** En = 75±5 going to 165

Fe_ei401_align_sym3.sqw
 1 ± 0.022 ; $J_0 = 28 \pm 0.3$; $J_1 = 0 \pm 0$; $\gamma = 31.0262 \pm$
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 7σ
 $\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

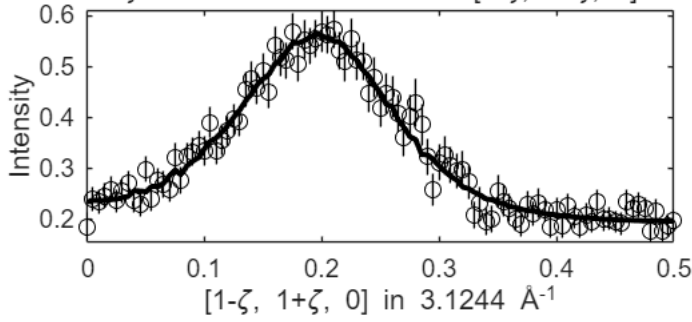


Fe_ei401_align_sym3.sqw
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 7σ
 $\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

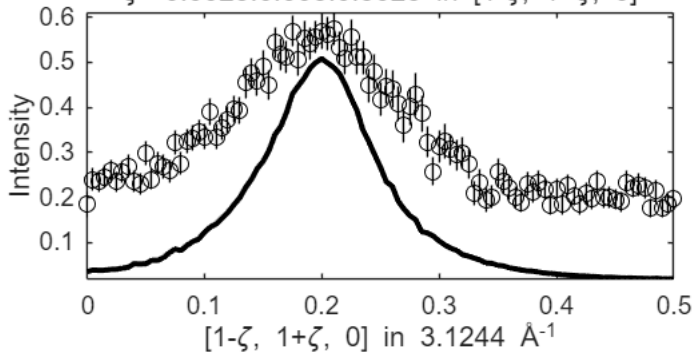


 **** En = 85±5 going to 165

Fe_ei401_align_sym3.sqw
 1 ± 0.022 ; $J_0 = 29 \pm 0.28$; $J_1 = 0 \pm 0$; $\gamma = 39.41 \pm 0.1$
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 80°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

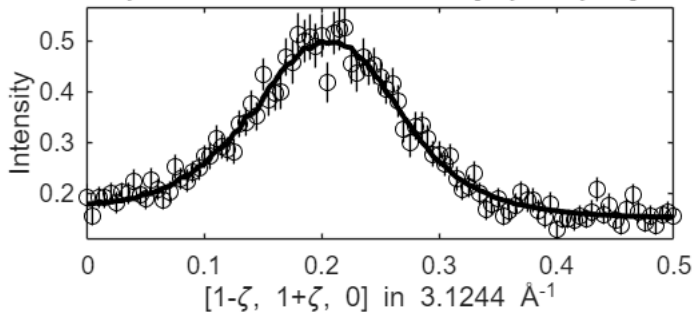


Fe_ei401_align_sym3.sqw
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 80°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

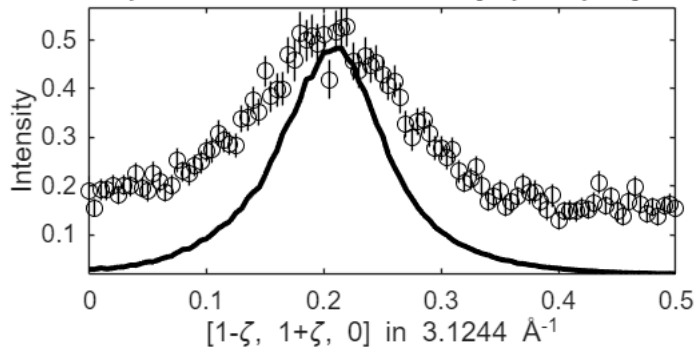


 **** $E_n = 95 \pm 5$ going to 165

Fe_ei401_align_sym3.sqw
 1 ± 0.023 ; $J_0 = 30 \pm 0.24$; $J_1 = 0 \pm 0$; $\gamma = 39.3005 \pm 0.1$
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 90°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

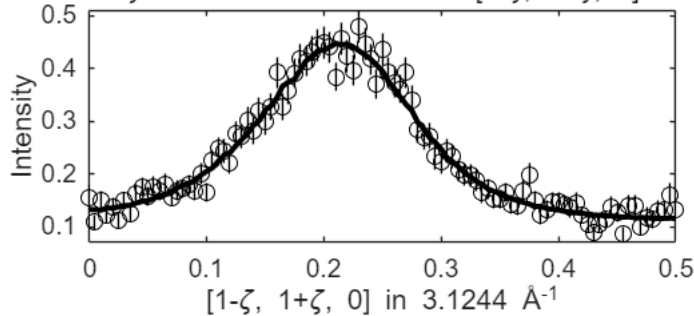


Fe_ei401_align_sym3.sqw
: 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 90
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

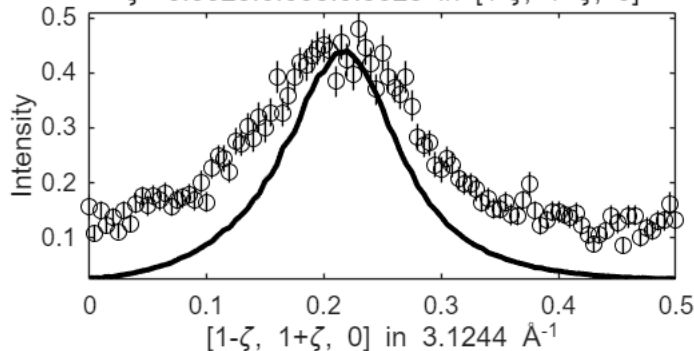


**** En = 105±5 going to 165

Fe_ei401_align_sym3.sqw
2±0.033; J0= 32±0.24; J1= 0± 0; gamma=49.3456±
0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 10
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

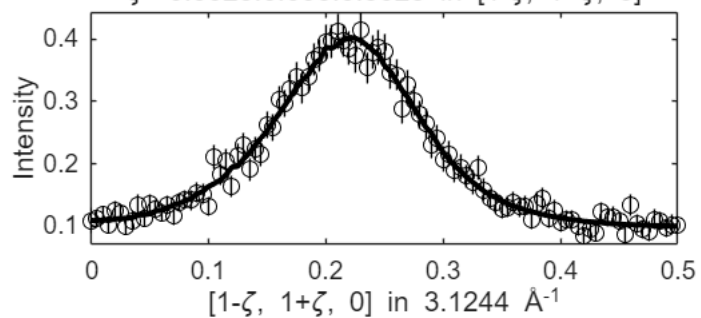


Fe_ei401_align_sym3.sqw
0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 10
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

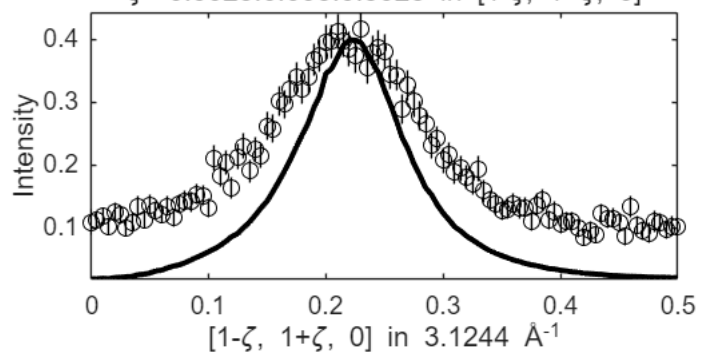


**** En = 115±5 going to 165

Fe_ei401_align_sym3.sqw
 1 ± 0.053 ; $J_0 = 33 \pm 0.38$; $J_1 = 0 \pm 0$; $\gamma = 45.0521 \pm 0.1$ in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, $11($
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

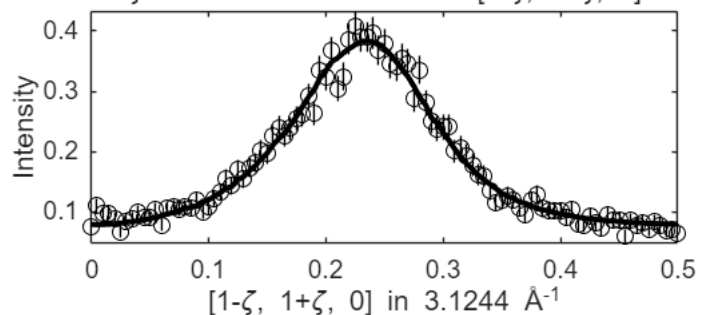


Fe_ei401_align_sym3.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, $11($
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

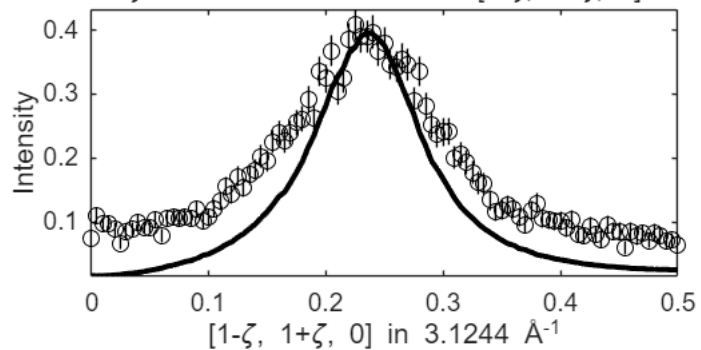


 **** $E_n = 125 \pm 5$ going to 165

Fe_ei401_align_sym3.sqw
 1 ± 0.049 ; $J_0 = 33 \pm 0.33$; $J_1 = 0 \pm 0$; $\gamma = 46.2692 \pm 0.1$ in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, $12($
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

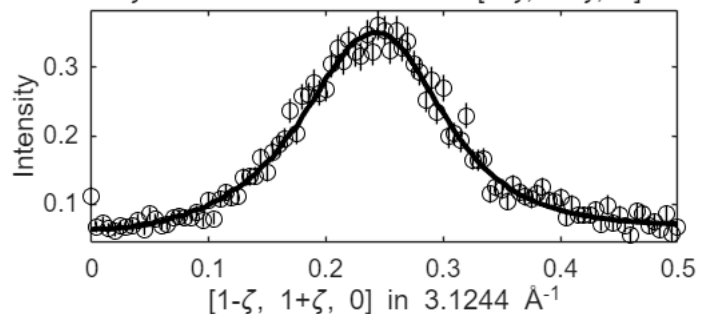


Fe_ei401_align_sym3.sqw
0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 120
 $\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

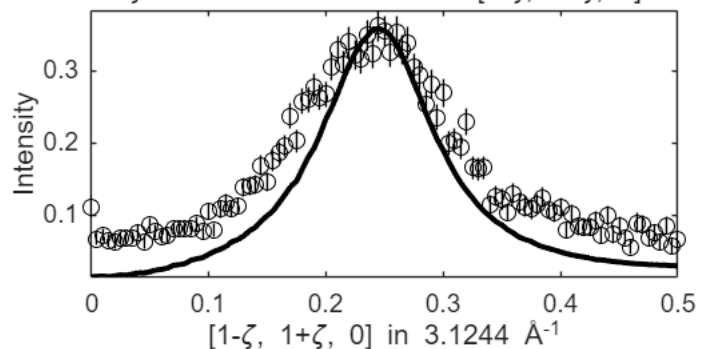


**** En = 135±5 going to 165

Fe_ei401_align_sym3.sqw
 1 ± 0.036 ; $J_0 = 34 \pm 0.21$; $J_1 = 0 \pm 0$; $\gamma = 50.8122 \pm$
0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 130
 $\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

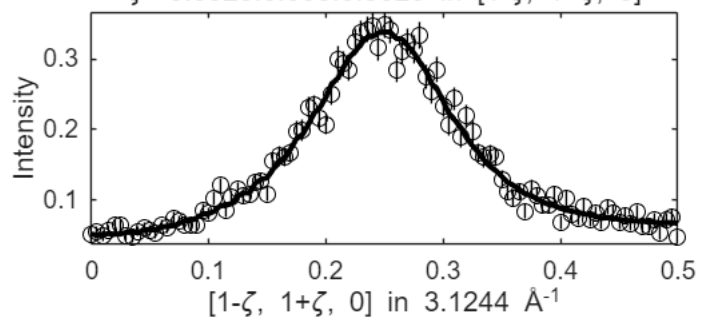


Fe_ei401_align_sym3.sqw
0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 130
 $\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

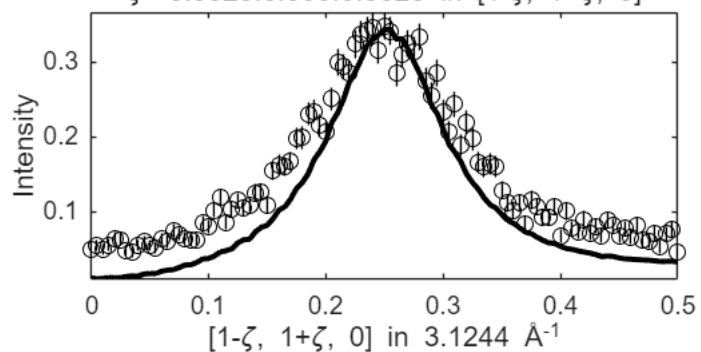


**** En = 145±5 going to 165

Fe_ei401_align_sym3.sqw
 1 ± 0.04 ; $J_0 = 35 \pm 0.24$; $J_1 = 0 \pm 0$; $\gamma = 53.9988 \pm 0.1$
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 140
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

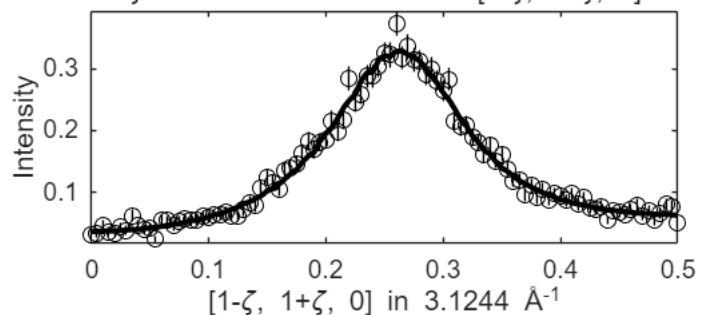


Fe_ei401_align_sym3.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 140
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

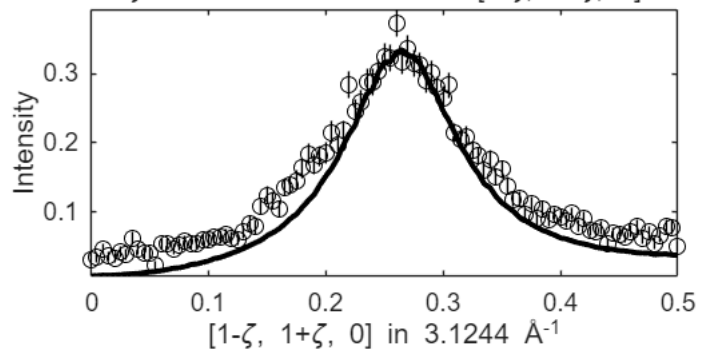


 **** $E_n = 155 \pm 5$ going to 165

Fe_ei401_align_sym3.sqw
 1 ± 0.037 ; $J_0 = 36 \pm 0.26$; $J_1 = 0 \pm 0$; $\gamma = 54.9047 \pm 0.1$
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 150
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

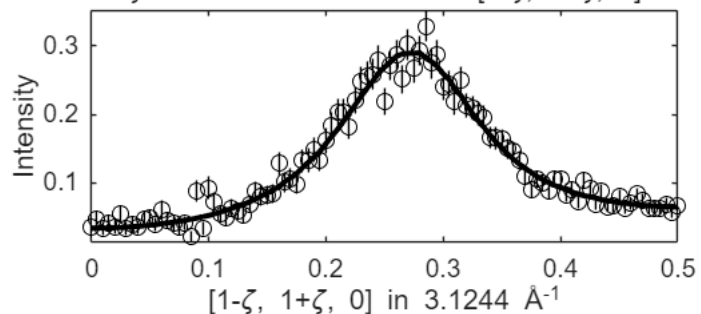


Fe_ei401_align_sym3.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 150
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

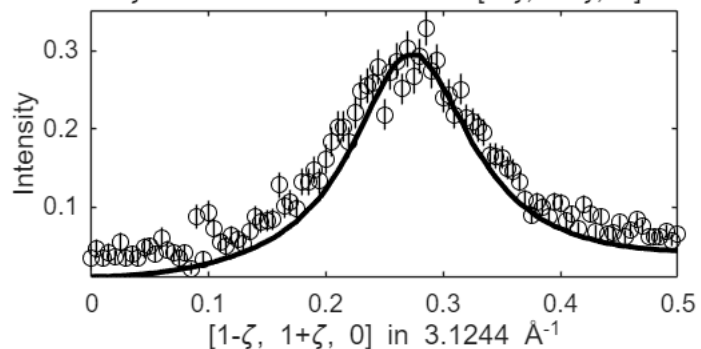


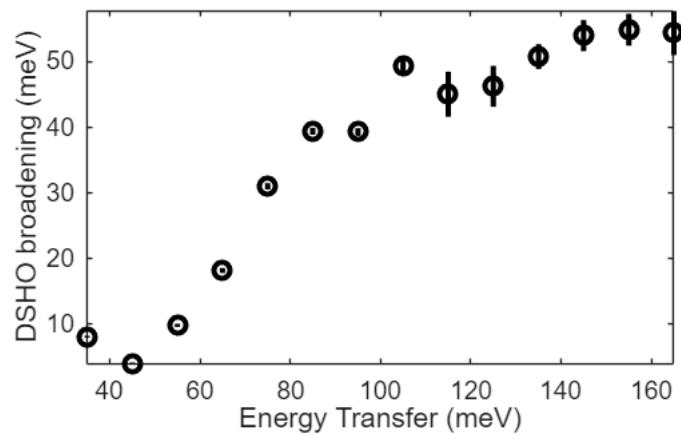
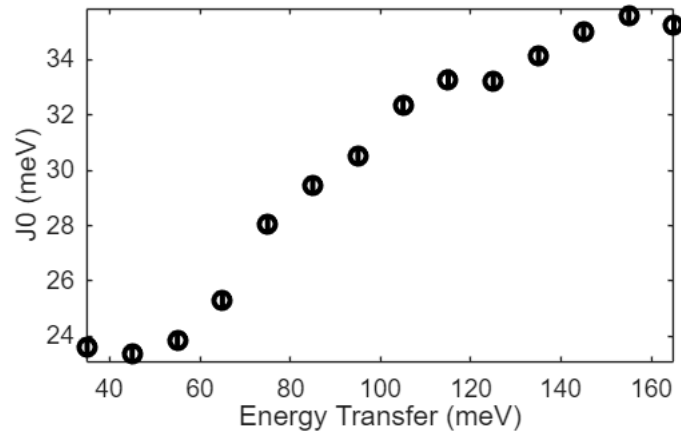
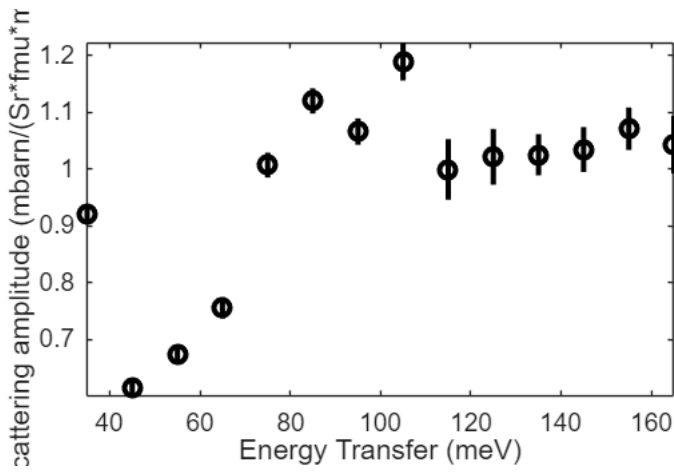
 **** En = 165±5 going to 165

Fe_ei401_align_sym3.sqw
 1 ± 0.051 ; $J_0 = 35 \pm 0.35$; $J_1 = 0 \pm 0$; $\gamma = 54.3885 \pm$
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 160
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



Fe_ei401_align_sym3.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 160
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$





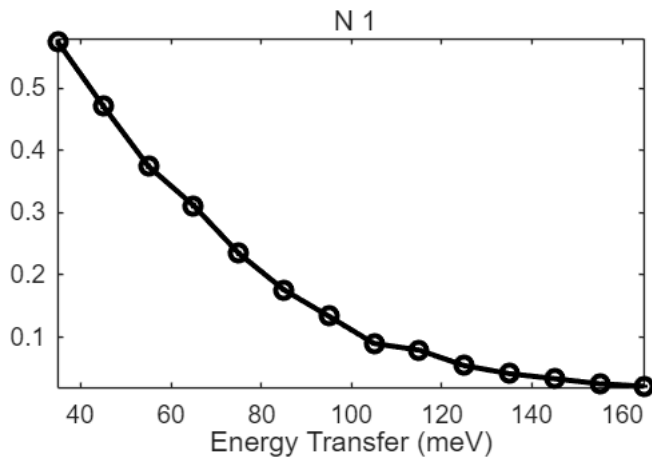
```

c1 = NG_FFCutsFits.(selectCuts);
NG_FFCutsFits.(selectCuts) = [c1{:}];

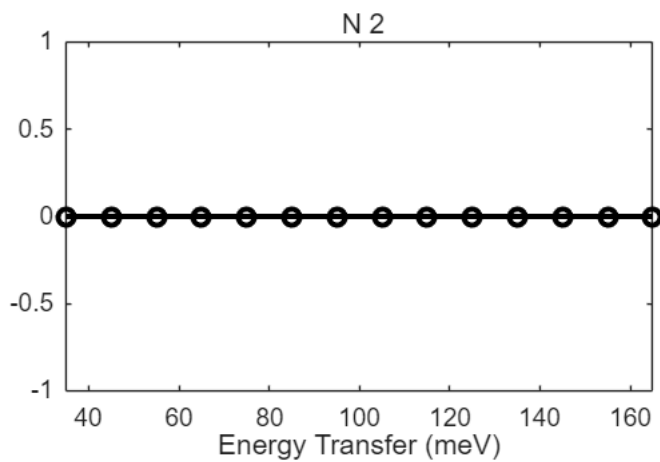
beep
beep
beep

all_fit_par = NG_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_NG.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_NG_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
fit_src_struc = NG_FFCutsFits;
save(Cutfitfile,"fit_src_struc","-v7.3')
```